ARQUITECTURAS E INFRAESTRUCTURAS PARA INTELIGENCIA ARTIFICIAL

DOCUMENTATION: HPC CLOUD SERVICES

Introduction:

Certain types of services related to high-performance computing are available in MS Azure. These services provide high-performance computing and GPU infrastructure with resource-rich clusters.

They also provide AI processing services such as **Azure Machine Learning.** This service allows you to allocate the necessary computing resources and make a payment for their use. It allows you to have from simple CPU and GPU resources, to compute clusters with a large number of nodes and memory:

https://azure.microsoft.com/es-es/products/machine-learning#tabx69c4fee3c8844375955db36c4f74da17

The AI services available in Azure consist of APIs that allow you to use these services to integrate them within applications.

https://learn.microsoft.com/es-es/azure/ai-services/

Let's look at an example that allows us to replace the YOLOv8 artificial intelligence model that we used to count people in an image with an Azure AI service. This is going to be achieved by using the following Computer Vision AI service for image labeling:

https://learn.microsoft.com/es-es/azure/ai-services/computer-vision/concept-tag-images-40

This service allows you to tag the objects that appear in the image.

Steps:

1) Create AI service

En Azure, busca el servicio “Azure AI services multi-service account”
Click en “Crear Azure AI services multi-service account”
Project Details:

- Subscription: Azure for Students
- Resource group: grupoest
- Instance details
  - Region: West Europe
  - Name: Enter a name
- Pricing plan: S0 (the free version (F0) is not available with this subscription.
- We tick the box to accept the terms and conditions.
- Click on "Next"
  - Type: All networks, including the internet, can access this resource
- Click on "Review and create"

- Click on "Create"

2) Access to the created resource

Go to "Keys and Endpoint". We will need to use one of the keys and the endpoint to access the service. This time, to facilitate the exercise, we are going to put this data directly in the code, but we should avoid putting keys in the code, and instead we would have to use some authentication and secret storage service, such as the Azure Key Vault service. https://learn.microsoft.com/en-us/azure/ai-services/use-key-vault?tabs=azure-cli&pivots=programming-language-python.

The following Python code snippet shows how to make an API call to tag it and get information from people who appear in it. This code will run on your local machine.

```
import os
from azure.ai.vision.imageanalysis import ImageAnalysisClient
from azure.ai.vision.imageanalysis.models import VisualFeatures
from azure.core.credentials import AzureKeyCredential

import json

# Set up credentials and client
endpoint = "ENDPOINT"
key = "KEY"

# Create an Image Analysis client for synchronous operations
client = ImageAnalysisClient(
    endpoint=endpoint,
    credential=AzureKeyCredential(key)
)

# Load image to analyze into a 'bytes' object
with open("frame.jpg", "rb") as f:
    image_data = f.read()


# Analyze image with PEOPLE feature
result = client.analyze(
    image_data=image_data,
    visual_features=[VisualFeatures.PEOPLE],
    language="en"
)

if result.people is not None:
        for person in result.people.list:
                confidence = person.confidence
                if confidence > 0.5:
                        bounding_box = person.bounding_box
                        print("Confidence:", confidence)
                        print("Bounding Box:", bounding_box)
                        print("Additional Information if available:")
                        print()
else:
    print("Analysis failed.")
    print("  Error reason:", result.error.reason)
    print("  Error code:", result.error.code)
    print("  Error message:", result.error.message)
```

3) Install the "azure-ai-vision" package on your local machine:

```
>pip3 install azure-ai-vision-imageanalysis
```

Then try running the code. Don't forget to look for a test image.

Extra Resources:

You may need to store and access files stored in your storage account when you do your internship. The first step is to manage application authentication in Azure. To do this, we

recommend following the Azure guide: "Authenticating applications hosted on Azure resources with the Azure SDK for Python."

https://learn.microsoft.com/es-es/azure/developer/python/sdk/authentication-azure-hosted-apps?toc=%2Fazure%2Fstorage%2Fblobs%2Ftoc.json&bc=%2Fazure%2Fstorage%2Fblobs%2Fbreadcrumb%2Ftoc.json&tabs=azure-portal%2Cazure-app-service

Finally, the Azure libraries must be used to read and write to the container. The following guide shows how to do it in Python.

https://learn.microsoft.com/en-us/azure/storage/blobs/storage-quickstart-blobs-python?tabs=managed-identity%2Croles-azure-portal%2Csign-in-azure-cli&pivots=blob-storage-quickstart-scratch

On the other hand, to send an image to the AI service from the serverless function, you must have it saved in a file and upload it using the function of the service's SDK. In the environment in which Azure functions run, the file system is read-only, except for the folder for temporary files. The following code snippet shows how to get the path to that folder:

```
# Create the full path to the file
tempFilePath = tempfile.gettempdir() # Obtain path to the temp. dir.
file_path = os.path.join(tempFilePath, "frame.jpg")
cv2.imwrite(file_path, frame)
vision_source = SDK. VisionSource(filename=file_path)
```